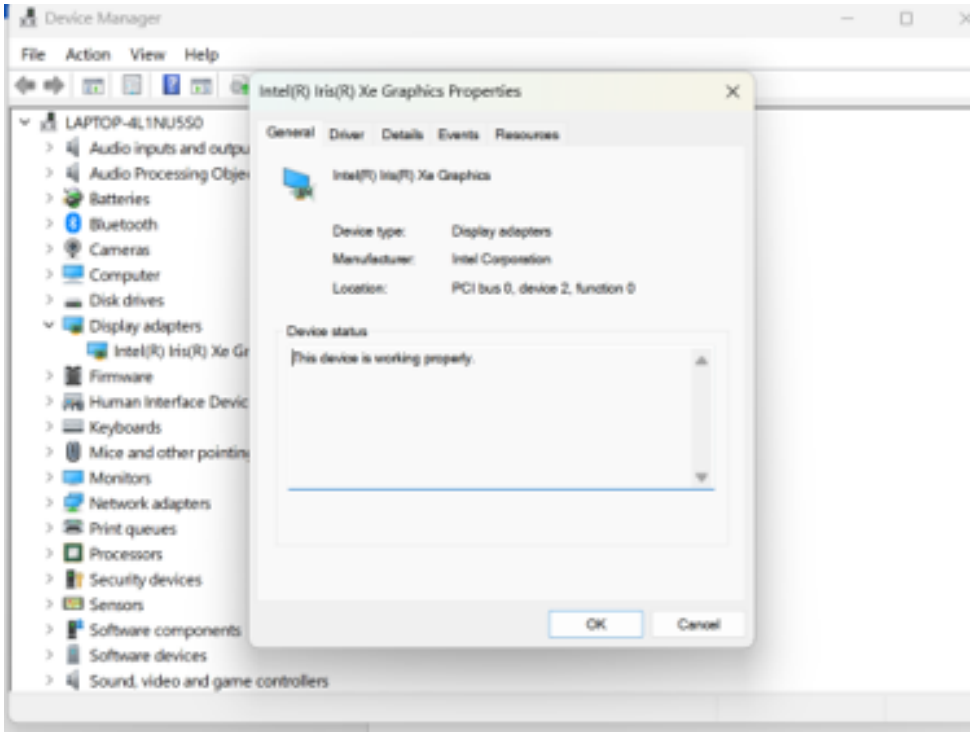
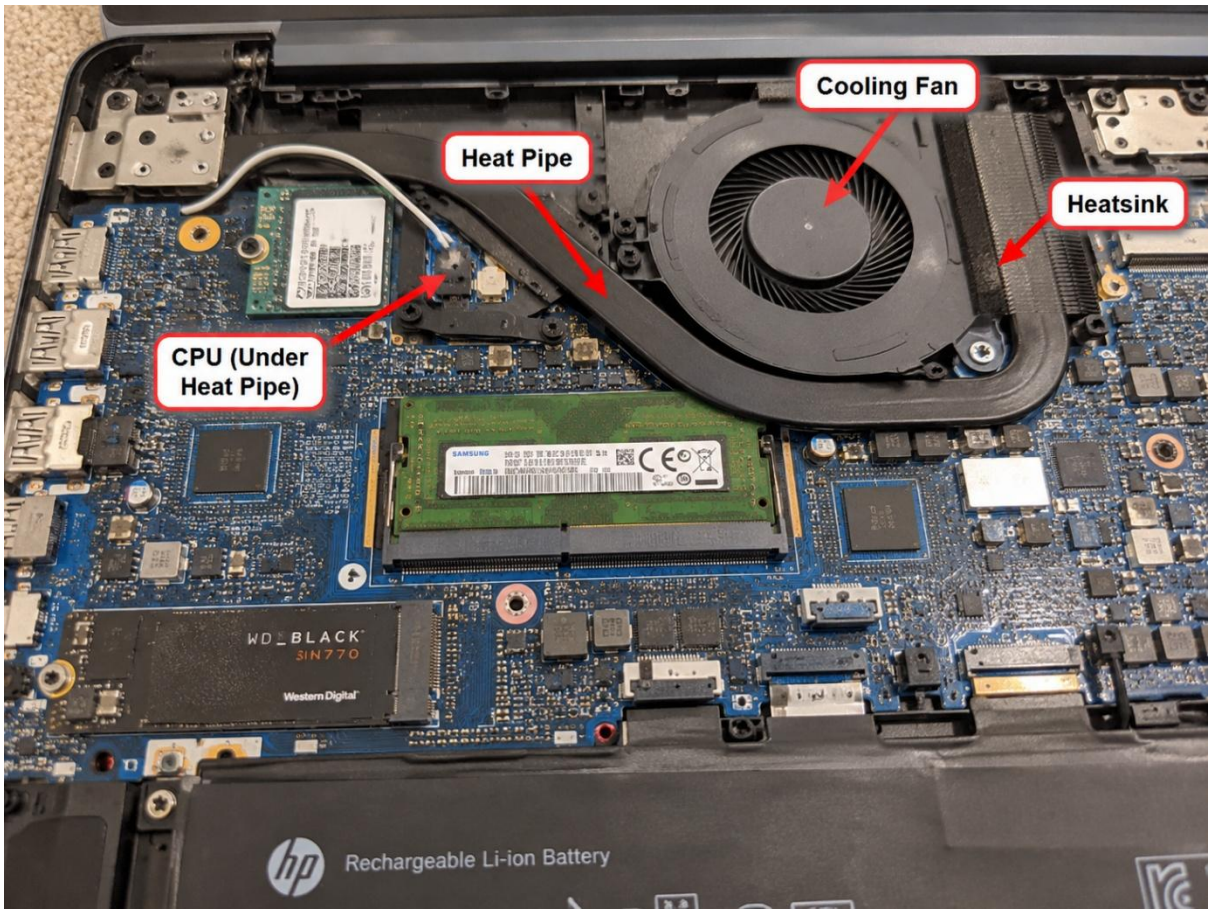
GPUs and Cooling Systems Theory

1.Open your own laptop or desktop (or use an online PC hardware simulator if you don't have access to a real system) and identify the GPU model installed. Write down the GPU's brand, model number, and whether it is integrated or dedicated.



Details	Information
GPU Brand	Intel
GPU Model	Intel(R) Iris(R) Xe Graphics
GPU Type	Integrated Graphics
Manufacturer	Intel Corporation
Device Status	Working Properly

2.Take a clear photo or screenshot of the cooling system (fan, heatsink, or liquid cooling setup) inside your PC or from a PC hardware simulator. Label the main cooling components and briefly describe their function



3.Create a comparison table listing at least 3 differences between air cooling and liquid cooling systems for GPUs, including factors like cost, efficiency, and maintenance.

Feature	Air Cooling	Liquid Cooling
Cooling Method	Uses fans and heatsinks to remove heat.	Uses liquid coolant to transfer heat away.
Cost	Generally cheaper.	Generally more expensive.
Cooling Efficiency	Good for normal gaming and daily use.	Generally better for high-performance GPUs.
Maintenance	Requires regular dust cleaning.	May require coolant maintenance.
Noise Level	Fans can produce more noise.	Can be quieter, but pump and radiator fans also make noise.
Installation	Easy to install.	More difficult to install.
Risk	Lower risk of leakage.	Liquid leakage can damage components.
Best Use Case	Normal PCs, office work and gaming.	High-performance gaming and heavy graphics workloads.

4.Imagine you are building a gaming PC for streaming IPL matches and playing graphics-heavy games. Choose between air cooling and liquid cooling for your GPU, and justify your choice in 3-4 sentences based on your research.

I would choose Liquid Cooling for my gaming PC because I plan to play graphics-heavy games and stream IPL matches at the same time. These activities put a heavy load on the GPU and generate more heat. Liquid cooling helps transfer heat away from the GPU efficiently, maintaining stable performance and reducing the chances of overheating during long sessions. Although it costs more than air cooling, I would prefer it for better cooling performance.